

Gate E3 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS** with finding X-27. Evidence directory: 20_EVIDENCE/E3/.

E3 — Formal conformance, static part (first pass, before any build)

Method: literal header extraction from the frozen source directory (identical to ZIP and manifest, E0) with the auditor's `extract_headers.py`; outputs `headers_defs.txt`, `headers_defs2.txt`, `headers_ext.txt`, `headers_chordal.txt`. Definitions opened manually: `RoutedSimplicialDefect`, `E32/Basic.lean`, `ExplicitFixedStability.lean`, `PublicationStability.lean`, `DefectExplicitPublication.lean`, `DefectSharpPublication.lean`, `E32/Consequences.lean`, `E32/Comparison.lean`, `E32/DeletionInduction.lean`, `E34/L11Main.lean`, `E34/L11Params.lean`, `E34/Params.lean`, `E34/Pinned.lean`, `RootPartitionStability`, `PieceEdgeStability`, `FixedDefectPieceEdges`, `OptimalTemplateObstruction`, `E35/Theorem.lean`, `E35/Main.lean`, `E18/Numeric.lean`, `MixedRounding/Defs.lean`, **annex** `BoundedCliqueGap/{Gap,Defs,AxiomCheck}.lean`. The elaborated types are NOT yet kernel-confirmed (E4 will `#check/#print` axioms them in the auditor build).

Definitions vs manuscript

Notion	Lean	Manuscript	Verdict
clique partition	<code>FarRounding.CliquePartition</code> : cliques, $\text{card} \geq 2$, pairwise edge-disjoint <code>pairs</code> , <code>biUnion pairs = edgeFinset</code>	exact edge partition (A.1)	MATCH
$M(n)$	<code>targetSize n = n * (n+1) / 6</code> ($\mathbb{N}$)	$\lfloor n(n+1)/6 \rfloor$	MATCH
$Q_s(n)$	<code>defectTarget s n = targetSize (n+s) - C(s+1, 2)</code> ($\mathbb{N}$ truncated)	$M(n+s) - C(s+1, 2)$	MATCH for n large (truncation only when $M(n+s) < C(s+1, 2)$)
$\text{rsd} \leq s$	<code>RoutedDefectAt</code> : $\forall U \subseteq R, R \subseteq U, R \text{ clique}, U \setminus R \neq \emptyset \rightarrow \exists v \in U \setminus R \exists C \subseteq N_U(v) \text{ clique}, N_U(v) \leq C + s$	§1.1.84	MATCH; <code>E32.rsd = Nat.find, rsd_le_ iff</code>
template $T(C,D,H)$	<code>defSplitGraph C D H</code> : $C \subseteq C, (C \cup D) \subseteq H$ adjacent, nothing else	$(K_{C \cup D}) \setminus V_{I_H}$	MATCH
root	<code>E32.IsDefectRoot</code> : disjoint cover, C clique of G , $ D =s$, $2 \leq C \leq H $	1.101, 1.899	MATCH
baseline	<code>rootBaseline = (C + D) H - C(C , 2)</code>	(6.10) $k(n-k) - C(k-s, 2)$	MATCH
canonical piece (defect)	<code>IsCanonicalPiece</code> : $\{x, h\}, x \in C \cup D, h \in H$ or $\{c, c', h\}$	1.908	MATCH
canonical piece (chordal)	<code>IsCanonicalAt</code> $R \subseteq K$: $ K \setminus R = 1 \wedge K \cap R \in \{1, 2\}$	1.868	MATCH
E_{bad}	Σ over noncanonical pieces of $C(K , 2)$	1.908	MATCH
optimal template	<code>IsAdmissibleExtrema</code> : $G = \text{defSplitGraph } C \ D \ H, D =s, \text{NearestCore}(n+s, C +s) = 6k - (2m+1) \leq 3$	"nearest integer to $(2(n+s)+1)/6$ "	MATCH
$\gamma_s, A_s, N_s^{\text{stab}}$	<code>stabilityGamma = min(min 1 (etaNF/4), epsS/4)/2</code> ; <code>stabilityConstant = max(40000(s+1)^3, 2/epsS)</code> ; <code>stabilityThreshold = 2 * deletionBase + sharpThreshold + s + 4</code> ; <code>deletionBase = minDegThreshold + F_s + s + 3 + ⌈1/ε_s⌉ + ⌈A_s⌉</code> ; <code>1 * minDegThreshold = ⌈10^50(s+1)^8⌉ + NLoc(NeditE, s, ε_s) + NfarE(etaNF s) + s + 5</code> ; <code>epsS = 1/(10^41(s+1)^8)</code>	(6.11), (F.1)	MATCH (term by term)
F.11 parameters	$\gamma_{11} = \lceil 256K^2/\varepsilon^2 \rceil$; $\delta_{11} = 1/(4\gamma^2)$; $B_{11} = \lfloor 8K/\delta_{11} \rfloor$; $m_{11}V = \lceil 2^{58}(K+1)^{15}/\varepsilon^{12} \rceil$	(F.11)	MATCH
pinned sample	<code>PinnedOn</code> : $\exists$ tree T , map ι preserving on-path relation (not required injective), subtrees $T_u \ni \iota(x_u)$, adjacency $\Leftrightarrow$ intersection	F.5	MATCH

tower	<code>E19.tower2.E35.Ptower s = cT(s+1)^1680·cT = hIter + cFar + 841780·cFar = cPoly(128·10^82)^105·cPoly = 4·17664^5·9000^105+3006·hIter = 4·8^5·2208^5·4500^105·(2·10^16)^105</code>	\$6.3, (6.28)	MATCH
s_max	<code>sMaxT n = Nat.findGreatest (tower2(Ptower S) ≤ n) n</code>	(6.29)	MATCH incl. the stated caveat

Headline theorems

All entries of E1's claim map were compared with the literal headers: quantifier order, exact coverage, order bounds, unrestricted lower bounds, real/rational deficits and the "root before Q and τ " order all match. See E1_RECORD.md for the table.

F.5 case structure (static)

`E34.lemma11_v = by_cases on (A) every admissible ψ has $\text{conf} \geq \delta_{11} n^2 \rightarrow \text{111_caseA} (\text{thm2} + \text{pinned} \Rightarrow \text{proper})$; else fix ψ with $\text{conf} < \delta_{11} n^2$; by_cases (B1) some section c has all rank assignments with $\text{mismatch} > \varepsilon/(2K) n^2 \rightarrow \text{111_caseB1} (\text{claim5_count}, \text{block_ineq}, \delta_{11} \cdot y^2 = 1/4, \text{prefix counting with } \text{card_prefix_le})$; else (B2) 111_caseB2 produces a chordal glued F with $d_E(G,F) < \varepsilon n^2$, contradicting farness. The three cases are exhaustive by construction (two nested by_cases), the common conclusion is "at most half of the n^m ordered samples are pinned". Matches Appendix F.5 text l.2320-2341.`

Export and target scope (static)

- `ReleaseExportCheck.lean`: only import `PaperIV`; exactly **224** #check lines (recount).
- 19 targets = `FREEZE_SCOPE` list; union of their import closures = all 607 modules (after correction C-01).
- `ResearchAudit` custom cone check over **75** literal targets (recount = manuscript's 75): forbids roots `Erdos81/GalvinRoute/RouteB` and any axiom outside `{propext, Classical.choice, Quot.sound}`.
- `OptimalTemplateObstructionAudit`: 6 targets, union cone, same checks.
- `FDCheck.ASCheck`: 6 selected declarations (public `Theorem C` eventual/maximum, `E35.theoremC.tower`, `fixed_defect_stability_explicit`, `fixed_defect_joint_stability_real`, `sequence_arbitrary_orders_stability`); forbids 6 named removal keys (`AlonShapira.lemma_4_2`, `two near_chordal_of_*` lemmas, `EditRoute`, `editApproxAt_all`, `EditRoute.fixedL4Localization_unconditional`, `AFKS.strong_regularity`). Scope = named lemmas only, as the manuscript states (§7 l.1138).
- Cone traversal code: type + value constants, recursive; axioms detected via `axiomInfo`. Limitation: does not detect relocated/renamed copies (stated in §7), and a constant missing from the environment is silently skipped (cannot happen for elaborated terms).

Observations

- E3-O1: E34 uses `AlonShapira.IsChordal / AlonShapira.editDist` (definitions in the `RequestProject` tree) — the manuscript discloses that shared definitions remain (§7). Their agreement with the paper's chordality is used through bridge lemmas; dynamic confirmation that the public types use `PaperIV.FarRounding.IsChordal / RootedDefectAt` only (they do, statically) is sufficient for the public statements.
- E3-O2: Several audit targets (`ResearchAudit`, `OptimalTemplateObstructionAudit`, `E17.ImprovedFarRoundingAudit`) contain no `#print axioms`; they rely on custom elaborators. E4 will not rely on them: the auditor writes an independent `#print axioms/cone` file.

E3 static status: PASS (no conformance defect found); dynamic part pending (E4).

Dynamic part (after E4, 2026-09-30 22:00 UTC)

- Elaborated types printed by `AuditorChecks.lean` agree with the static headers; all 50 listed public declarations exist in the environment built from the cut; 224 exports compile through `import PaperIV` alone.
- The 19 targets are exactly `FREEZE_SCOPE`; their closure is 607 modules (runner `PLAN.json`).
- Cone checks: standard axioms only; no `Erdos81` root; `ASCheck` exclusions confirmed on the six named cones.
- **X-27 (MODERATE)**: `E32.cp_classification_of_theoremCPrimeC` (cited in Table 9 for the unrestricted equality classification) and `E32.ref15Theorem11_of_theoremCPrimeC` depend, via `E32.theoremC_at` $\rightarrow$

`DefectSharpPublication.rooted_defect_eventual`, on the historical Alon–Shapira removal chain (`AlonShapira.lemma_4_2`, `AFKS.strong_regularity`, `EditRoute.fixedL4Localization_unconditional`, ...). The manuscript presents that chain as historical and not selected (§7 l.1138, E.2 l.1951, ref. [23]) without saying that the cp-part of (6.18) and the formal comparison with [15, Theorem 1.1] still use it.

E3 verdict: PASS with one MODERATE provenance finding (X-27). No statement–type mismatch.

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